Home Lab 8: Pen and Marker Chromatography

Do this with a parent. Goggles on. Clean up after.

Goal

Separate the dyes in different markers and compute Rf, then match a "mystery note" to a marker.

You need

From the kit: coffee-filter paper (or cut strips from a paper towel). From home: water, a few clear cups, pencil, a ruler, tape, and **3–4 different water-soluble markers** (washable markers or non-permanent felt pens — **black** ones separate best). Have a parent secretly write a short "mystery note" with one of the markers on a strip and label it "Note."

Steps

1. Cut strips about 2 cm wide and as tall as your cup.
2. On each strip, draw a **pencil** line 1.5 cm from the bottom. Label the top in pencil (marker color/brand).
3. Put a small dot of one marker on the pencil line of its strip. For the "Note" strip, dot it with ink scraped from the note, or re-trace a bit of the note right on the line.
4. Pour ~1 cm of water into each cup. Tape each strip so it hangs with the bottom edge in the water but the **pencil line above the water**.
5. Wait. When the water climbs to about ¾ of the strip, take it out and **immediately** mark the water line (the solvent front) in pencil.
6. Let it dry. Mark the center of each separated color band.
7. Measure, from the pencil start line: to the solvent front, and to each color band.
8. **Rf = distance the color moved ÷ distance the solvent front moved.**

Results

Solvent front distance: ______ cm

Strip	Color band	Distance moved	Rf
Marker A			
Marker A			
Marker B			
Note			
Note			

Follow-up

1. Which marker's strip has the same colors at the same Rf values as the "Note" strip?
2. Which color in your favorite marker traveled farthest? Is it more or less soluble in water than the ones that stayed near the bottom?
3. Why did you use pencil and not pen for the lines?
4. Can an Rf be more than 1? Why not?